Talaksh Rabari

M.Tech – ICT (Software Systems)

talakshrabari2017@gmail.com linkedin github

Education

Dhirubhai Ambani University (DAU / DA-IICT) 2025 – Present
M.Tech in Software Systems (CPI: 8.66 / 10) *Gandhinagar, Gujarat*

Sardar Vallabhbhai Patel Institute of Technology (SVIT) 2021 – 2025
B.E. in Information Technology (CPI: 8.43 / 10) *Vasad, Gujarat*

Experience

Dhirubhai Ambani University July 2026 – Present
Teaching Assistant – Information Retrieval *Gandhinagar, Gujarat*

- Assisted 50+ students in understanding core search concepts including text tokenization, TF-IDF, Vector Space Models, and Cosine Similarity.
- Helped students understand the connection between classical information retrieval and modern vector search used in RAG systems.

H P Param IT Solutions Feb 2025 – June 2025
FrontEnd Developer Intern *Anand, Gujarat*

- Developed modular, responsive cross-platform web interfaces using ReactJS, Tailwind CSS, and modern hooks (`useState`, `useEffect`, Context API).
- Integrated frontend components with backend RESTful APIs, implemented client-side state handling, and conducted cross-browser testing to optimize rendering performance.
- Collaborated via **Git** branch workflows, participated in pull request reviews, and resolved critical production UI bugs.

Projects

GSSTB Scholar (Agentic RAG Platform) | *LangGraph, FastAPI, ChromaDB, React 18, Groq, SSE* GitHub | Live Demo

- Architected an autonomous Agentic RAG pipeline with LangGraph and Groq (Llama-3.3-70B), implementing query contextualization, document grading, and a self-correcting hallucination-check loop.
- Engineered a Hybrid Search system fusing ChromaDB vector search and BM25 sparse search via Reciprocal Rank Fusion (RRF, $k = 60$) and Cross-Encoder reranking.
- Built a dual-engine document ingestion pipeline (PyMuPDF + EasyOCR) and exposed asynchronous FastAPI REST endpoints using Server-Sent Events (SSE) to stream responses to a React 18 frontend.

AlumNet (Alumni-Student Portal) | *React.js, Firebase Auth, Firestore, Gemini API, Tailwind CSS* GitHub | Live Demo

- Engineered a secure role-based full-stack web application for Students, Alumni, and Admins utilizing Firebase Authentication and granular Firestore Security Rules.
- Developed modules for Alumni Directory, Referral Tracking, Job Board, Direct Messaging, and an Admin approval workflow.
- Integrated Gemini API for AI-driven automated resume parsing and intelligent career path recommendations.

Cross-Lingual Information Retrieval Pipeline | *Python, PyTorch, FAISS IVFPQ, HuggingFace, BM25* GitHub

- Engineered a scalable ingestion pipeline indexing 8.8M passages using FAISS IVFPQ vector quantization for high-throughput, low-latency search.
- Benchmarked dense bi-encoders against lexical baselines across 5 Indic languages, evaluating NDCG@10 and MRR@10 on TREC-DL benchmarks for RAG pipeline optimization.

Technical Skills

Languages: Python, JavaScript, SQL, C++
Frontend: React.js, Next.js (Basics), Tailwind CSS, HTML5/CSS3, Modular UI Architecture
Backend & APIs: FastAPI, Node.js,RESTful APIs, Server-Sent Events (SSE)
Databases & Cloud: PostgreSQL, Firebase/Firestore, ChromaDB
DevOps & Tools: **Docker**, Git, GitHub Actions (CI/CD), VS Code
AI / LLM Engineering: LangGraph, Agentic Workflows, RAG Architectures, Vector Search (FAISS), Embeddings

Leadership & Achievements

- **Teaching Assistant (Intro to ICT, DBMS):** Evaluated coursework submissions, mentored undergraduates, and designed lab exercises. *July 2025 – Dec 2025*
- **GATE CS-IT 2025:** Qualified with a Score of **518**.
- **Winter of Code 2025-26:** Secured **2nd Place** in Mid-Evaluation organized by DAU MSTC Club.